Discrete Structures for Computer Science

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Lecture #11: Integers and Modular Arithmetic



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Based on materials developed by Dr. Adam Lee



Today's Topics

Integers and division

- The division algorithm
- Modular arithmetic
- Applications of modular arithmetic



What is number theory?

Number theory is the branch of mathematics that explores the integers and their properties.

Number theory has many applications within computer science, including:

- Organizing data
- Encrypting sensitive data
- Developing error correcting codes
- Generating “random” numbers
- …

We will only scratch the surface…

The notion of divisibility is one of the most basic properties of the integers



Definition: If a and b are integers and $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$. We write $a \mid b$ to say that a divides b, and $a \nmid b$ to say that a does not divide b.

Mathematically: $a \mid b \Leftrightarrow \exists\, c \in \mathbf{Z}\ (b = ac)$

Note: If $a \mid b$, then

- a is called a factor of b
- b is called a multiple of a

We've been using the notion of divisibility all along!

- $E = \{x \mid x = 2k \wedge k \in \mathbf{Z}\}$



Division examples

Examples:

- Does 4 | 16?
- Does 3 | 11?
- Does 7 | 42?

Question: Let n and d be two positive integers. How many positive integers not exceeding n are divisible by d?

Answer: We want to count the number of integers of the form dk that are less than n. That is, we want to know the number of integers k with $0 \leq dk \leq n$, or $0 \leq k \leq n/d$. Therefore, there are $\lfloor n/d \rfloor$ positive integers not exceeding n that are divisible by d.



Important properties of divisibility

Property 1: If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$

Property 2: If $a \mid b$, then $a \mid bc$ for all integers c.

Property 3: If $a \mid b$ and $b \mid c$, then $a \mid c$.



Division algorithm

Theorem: Let a be an integer and let d be a positive integer. There are unique integers q and r, with $0 \leq r < d$, such that $a = dq + r$.

For historical reasons, the above theorem is called the division algorithm, even though it isn't an algorithm!

Terminology: Given $a = dq + r$

- a is called the dividend
- d is called the divisor
- q is called the quotient
- r is called the remainder
- $q = a$ **div** d
- $r = a$ **mod** d

div and mod are operators



Examples

Question: What are the quotient and remainder when 123 is divided by 23?

Answer: We have that $123 = 23 \times 5 + 8$. So the quotient is 123 **div** 23 = 5, and the remainder is 123 **mod** 23 = 8.

Question: What are the quotient and remainder when -11 is divided by 3?

Answer: Since $-11 = 3 \times -4 + 1$, we have that the quotient is -4 and the remainder is 1.

Recall that since the remainder must be non-negative, $3 \times -3 - 2$ is not a valid use of the division theorem!

Many programming languages use the **div** and **mod** operations



For example, in Java, C, and C++

- / corresponds to **div** when used on integer arguments
- % corresponds to **mod**

```
public static void main(String[] args)
{
    int x = 2;
    int y = 5;
    float z = 2.0;

    System.out.println(y/x);
    System.out.println(y%x);
    System.out.println(y/z);
}
```

Prints out 1

Prints out 2, not 2.5!

Prints out 2.5

This can be a source of many errors, so be careful in your future classes!

In-class exercises

Problem 1 & 2: On Top Hat

Problem 3: Show that if a is an integer and d is an integer greater than 1, then the quotient and remainder obtained dividing a by d are $\left\lfloor \frac{a}{d} \right\rfloor$ and $a - d\left\lfloor \frac{a}{d} \right\rfloor$, respectively.

Sometimes, we care only about the remainder of an integer after it is divided by some other integer

Example: What time will it be 22 hours from now?



Answer: If it is 6am now, it will be (6 + 22) **mod** 24 = 28 **mod** 24 = 4 am in 22 hours.

Since remainders can be so important, they have their own special notation!

Definition: If a and b are integers and m is a positive integer, we say that a is congruent to b modulo m if $m \mid (a - b)$. We write this as $a \equiv b \pmod{m}$.

Note: $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$.

Examples:

- Is 17 congruent to 5 modulo 6?
- Is 24 congruent to 14 modulo 6?



Properties of congruencies

Theorem: Let m be a positive integer. The integers a and b are congruent modulo m ($a \equiv b \pmod{m}$) iff there is an integer k such that $a = b + km$.

Theorem: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- $(a + c) \equiv (b + d) \pmod{m}$
- $ac \equiv bd \pmod{m}$

Congruencies have many applications within computer science



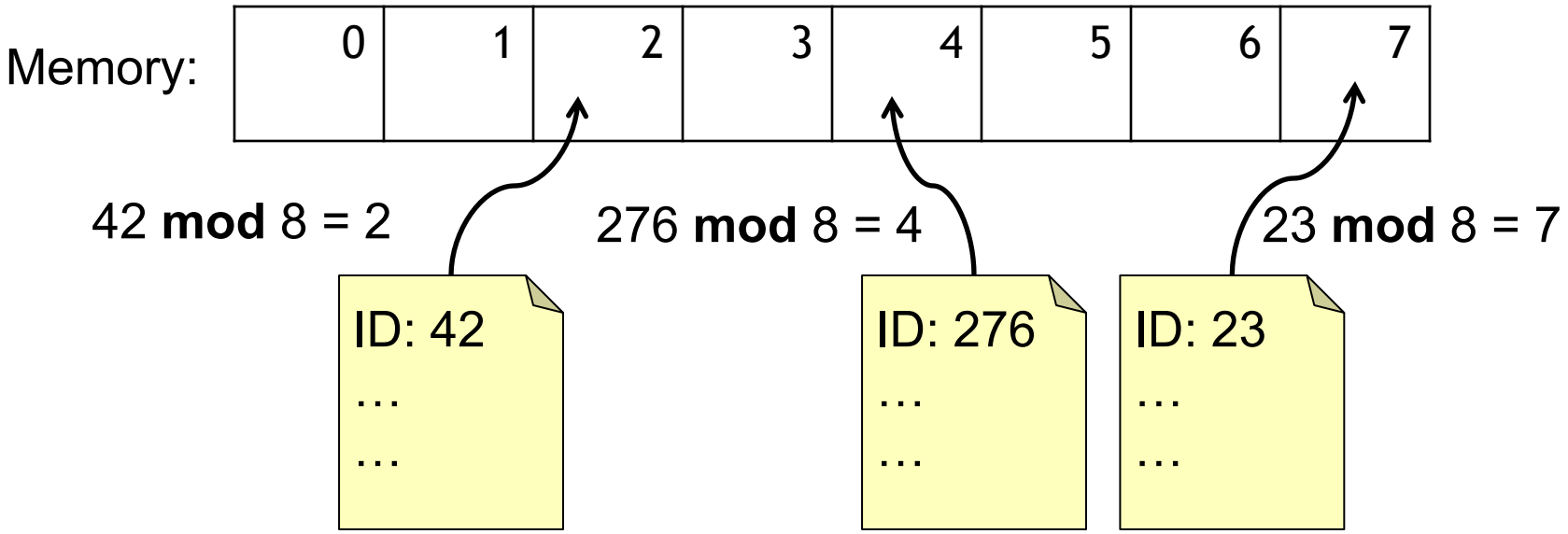
Today we'll look at three:

1. Hash functions
2. The generation of pseudorandom numbers
3. Cryptography

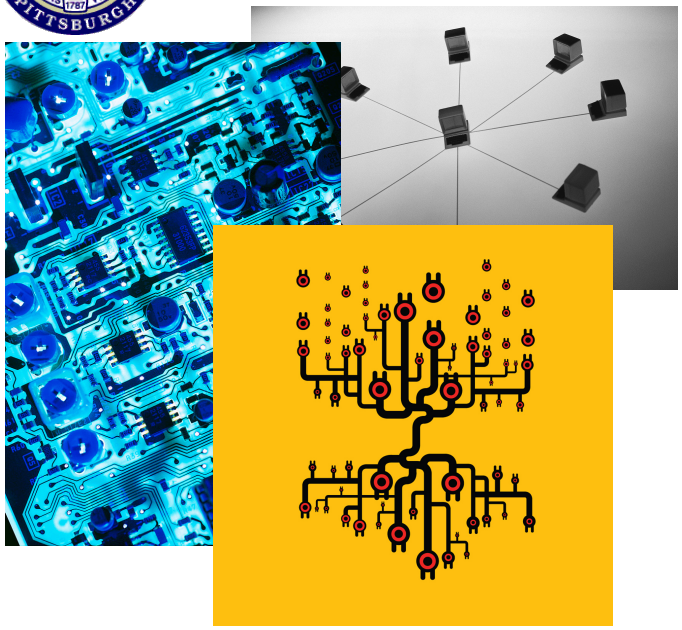
Hash functions allow us to quickly and efficiently locate data

Problem: Given a large collection of records, how can we find the one we want quickly?

Solution: Apply a hash function that determines the storage location of the record based on the record's ID. A common hash function is $h(k) = k$ **mod** n, where n is the number of available storage locations.



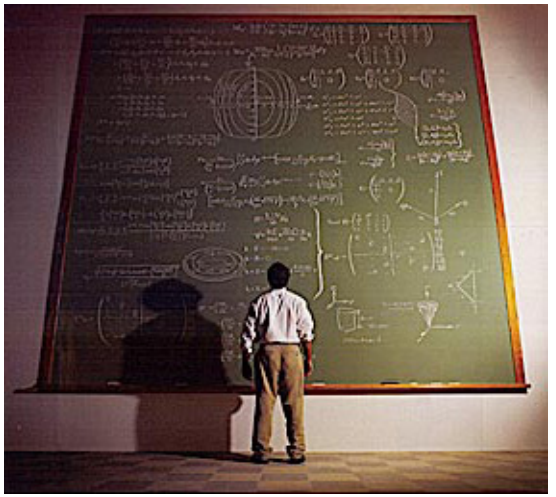
Many areas of computer science rely on the ability to generate pseudorandom numbers



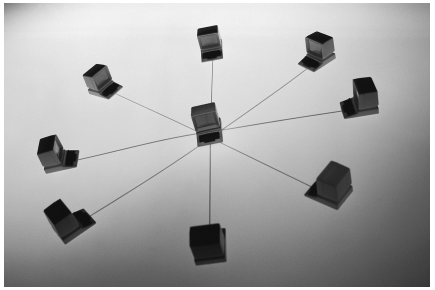
Hardware, software, and network simulation



Security



Coding algorithms



Network protocols

Congruencies can be used to generate pseudorandom sequences



Step 1: Choose

- A modulus m
- A multiplier a
- An increment c
- A seed x_0

Step 2: Apply the following

- $x_{n+1} = (ax_n + c) \bmod m$

Example: $m = 9$, $a = 7$, $c = 4$, $x_0 = 3$

- $x_1 = 7x_0 + 4 \bmod 9 = 7 \times 3 + 4 \bmod 9 = 25 \bmod 9 = 7$
- $x_2 = 7x_1 + 4 \bmod 9 = 7 \times 7 + 4 \bmod 9 = 53 \bmod 9 = 8$
- $x_3 = 7x_2 + 4 \bmod 9 = 7 \times 8 + 4 \bmod 9 = 60 \bmod 9 = 6$
- $x_4 = 7x_3 + 4 \bmod 9 = 7 \times 6 + 4 \bmod 9 = 46 \bmod 9 = 1$
- $x_5 = 7x_4 + 4 \bmod 9 = 7 \times 1 + 4 \bmod 9 = 11 \bmod 9 = 2$
- …

The field of cryptography makes heavy use of number theory and congruencies



Cryptography is the study of secret messages

Uses of cryptography:

- Protecting medical records
- Storing and transmitting military secrets
- Secure web browsing
- …

Congruencies are used in cryptosystems from antiquity, as well as in modern-day algorithms

Since modern algorithms require quite a bit of background to discuss, we'll examine an ancient cryptosystem

The Caesar cipher is based on congruencies



To encode a message using the Caesar cipher:

- Choose a shift index s
- Convert each letter A-Z into a number 0-25
- Compute $f(p) = p + s$ **mod** 26

Example: Let $s = 9$. Encode "ATTACK".

- ATTACK = 0 19 19 0 2 10
- $f(0) = 9$, $f(19) = 2$, $f(2) = 11$, $f(10) = 19$
- Encrypted message: 9 2 2 9 11 19 = JCCJLT

Decryption involves using the inverse function



That is, $f^{-1}(p) = p - s \bmod 26$

Example: Assume that $s = 3$. Decrypt the message "UHWUHDW".

- UHWUHDW = 20 7 22 20 7 3 22
- $f^{-1}(20) = 17$, $f^{-1}(7) = 4$, $f^{-1}(22) = 19$, $f^{-1}(3) = 0$
- Decrypted result: 17 4 19 17 4 0 19 = RETREAT



In-class exercises

Problem 4 & 5: On Top Hat



Final thoughts

- Number theory is the study of integers and their properties
- Divisibility, modular arithmetic, and congruency are used throughout computer science
- Next time:
  - Prime numbers, GCDs, integer representation (Section 4.2 and 4.3)